Shah Nawaz Haider

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Education

University of Science and Technology Chittagong (USTC) — *Chittagong, Bangladesh*
B.Sc. in Computer Science & Engineering. CGPA: 3.72/4.00 — May 2026

Thesis: *QuantFlow: A Federated Mamba-Based Post-Transformer Foundation Model for Time-Series Forecasting*
Advisor: Dr. Hadaate Ullah

Research Interests

Foundation Models, World Models, Large Language Models, Agentic Reasoning, Multimodal Learning, Interpretability

Highlights

- 13 peer-reviewed / preprint publications, including 2 as first author.
- Research Supervisor, ResearchBuddy AI, leading two applied NLP research teams.
- Reviewer, *SN Computer Science* (Springer Nature).
- Industry experience building production LLM and ML systems for financial risk modeling at scale.

Research & Industry Experience

Synnax AI — *Dubai, UAE (Remote)*
Data Scientist — Aug 2024 – May 2026

- Engineered end-to-end ML pipelines for credit risk and default prediction over SEC filings and market data spanning 16,000+ public companies.
- Built LLM-powered agents to extract risk signals from 10-K, 10-Q, and 8-K filings, automating detection of high-risk and defaulted entities.
- Converted large-scale unstructured regulatory data into production-ready features within a production MLOps stack.

ResearchBuddy AI (Private Research Lab)
Research Supervisor — Jul 2026 – Present

- Lead two research teams focused on Bengali Dialect Corpus Enhancement and Edge-Efficient Embedding Models.
- Direct research strategy, oversee benchmarking, and coordinate publication efforts across both teams.

Quantropy — *Chittagong, Bangladesh*
AI/ML Instructor — Jun 2026 – Present

- Designed and taught an AI curriculum covering deep learning, LLMs, foundation models, and modern generative AI.
- Ran hands-on workshops on transformers, retrieval-augmented generation (RAG), and AI agents.

Omdena — *Milan, Italy (Remote)*
Data Scientist (Part-time) — May 2024 – Jul 2024

- Built and evaluated ML models for crop optimization and resource allocation from environmental and spatial agricultural datasets.

University of Science and Technology Chittagong (USTC) — *Chittagong, Bangladesh*
Peer Tutor (Part-time) — Sep 2023 – Nov 2023

- Mentored undergraduates in programming, algorithms, and problem-solving within introductory computer science courses.

Selected Publications

First-Author

[1] **Haider, S. N.**, Austin, S., Barua, A., Shawon, S. M., & Ullah, H. QuantFlow: A Federated Mamba-Based Post-Transformer Foundation Model for Time-Series Forecasting. *arXiv:2607.02632* (Preprint).

[2] **Haider, S. N.**, Barua, A., Austin, S., & Shawon, S. M. Lightweight Federated and Explainable AI-Enhanced Crop Yield Forecasting in Heterogeneous Agricultural Systems. *IEEE ICCIT*, 2025. DOI

Selected Co-Authored

[1] Shawon, S. M., **Haider, S. N.**, Barua, A., Austin, S., Adan, I. A., Hossain, M. S., & Zubair, H. T. Hybrid CNN-LSTM Model for Urban Energy Load Forecasting with IGA-XAI for Smart Grids. *Results in Engineering* (Elsevier, Q1), 2025. DOI

[2] Barua, A., **Haider, S. N.**, Austin, S., & Shawon, S. M. Urban Energy Load Dataset of Chattogram: Peak and Off-Peak Variability Insights. *Data in Brief* (Elsevier), 2025. DOI

[3] Shawon, S. M., Dutta, P., **Haider, S. N.**, Barua, A., Austin, S., & Adan, I. A. Agri-Light-Fed: Lightweight Federated Learning with Explainable AI for Efficient and Privacy-Preserving Crop Yield Prediction. *IEEE ICCIT*, 2025. DOI

[4] Shawon, S. M., **Haider, S. N.**, Barua, A., Austin, S., & Noor, K. R. Predictive Modelling for Multi-Location Deep Learning-Based Load Forecasting: An Integrated Approach. *IEEE ICECE*, 2024. DOI

[5] Austin, S., **Haider, S. N.**, Shil, R., Lija, N. S., & Noor, K. R. Bi-LSTM-RF Based Hybrid Deep Learning Approach to Identify Level of Depression from Bengali Facebook Status. *IEEE ICCIT*, 2024. DOI

[6] Shawon, S. M., **Haider, S. N.**, Chakma, A., Alam, M. W., Islam, M. T., & Rana, M. F. DeepFlowNet: Deep Learning-Based Daily Water Flow Forecasting of Test River. *IEEE PEEIACON*, 2024. DOI

[7] Shawon, S. M., **Haider, S. N.**, Austin, S., & Barua, A. HybridNet: ResNet50-GRU Integration for Enhanced Rice Leaf Disease Classification in Precision Agriculture. *Springer LNNS (MIET)*, 2024. DOI

Complete publication list and citation metrics available on Google Scholar.

Research Projects

Bengali BPE Tokenizer *GitHub*

- Trained a Byte Pair Encoding tokenizer from scratch for Bengali and other non-Latin scripts, achieving 7.33× fewer tokens than GPT-2 across 3.4 billion characters of the CC-100 Bengali corpus.
- Implemented custom multi-byte UTF-8 and conjunct handling, cutting token inflation and inference cost for under-resourced languages.

Lunar Seismic Data Visualization - NASA International Space Apps Challenge *GitHub*

- Built an interactive web app to visualize lunar seismic data from NASA's Planetary Data System, overlaying mineral composition and Bouguer gravity disturbances against seismic events.

Professional Service

- Peer Reviewer, *SN Computer Science*, Springer Nature

Awards

- **Champion, Project Competition** - Research Colloquium, USTC. Chittagong, Bangladesh *May 2026*
- **Champion, Poster Competition** - Winter School Colloquium, USTC. Chittagong, Bangladesh *February 2025*
- **Regional Champion** - NASA International Space Apps Challenge. Dhaka, Bangladesh *October 2023*

Technical Skills

Programming	Python, Java, JavaScript, C/C++, MATLAB, SQL
ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers
LLM & Agentic AI	LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, LoRA/QLoRA
Data & Databases	Pandas, NumPy, Matplotlib, Plotly, PostgreSQL, MongoDB, Redis

Leadership & Service

Programming Club, USTC *Chittagong, Bangladesh*
President Jul 2024 – Jun 2025

- Led the university programming club, organizing technical workshops and coding competitions for CS students.

Robotics Society, USTC *Chittagong, Bangladesh*
Member Jul 2023 – Jun 2025

- Architected an autonomous path-following robotic system by optimizing sensor fusion algorithms.

Certifications

- **Machine Learning Specialization**, Stanford University (Coursera) *Dec 2023*
- **Data Analysis with Python**, freeCodeCamp *Mar 2023*